WWN Zoning (World Wide Name Zoning)

Zoning
In a **Fibre Channel (FC) switch**, zoning is a function that **logically segments node ports within a SAN fabric into groups**.

- Devices (initiators and targets) within the **same zone** can communicate with each other.
- Devices in **different zones** cannot communicate unless explicitly allowed.

This segmentation helps **improve security, reduce unnecessary traffic, and simplify management** in a SAN environment.

- **Definition:** Uses the **WWN** (unique identifier) of the HBA (initiator) and storage port (target) to control access.
- **How it works:**
    - Each device in a SAN has a unique 64-bit WWN.
    - Zoning rules are created based on these WWNs instead of physical switch ports.
- **Advantages:**
    - Cabling changes do **not** affect zoning (because it's based on device identity, not port).
    - Flexible and easy to maintain in growing SANs.
- **Disadvantages:**
    - Requires correct WWN mapping — more work when adding devices.

2. Port Zoning

- **Definition:** Uses the **physical switch port number** where the device is connected to control access.
- **How it works:**
    - The zoning rule applies to the **port** on the SAN switch, not the device itself.
    - If a device is moved to another port, zoning must be updated.
- **Advantages:**
    - Simple and easy to configure.
    - Good for environments where devices rarely move.
- **Disadvantages:**
    - Any change in cabling requires reconfiguration of zoning.

3. Mixed Zoning

- **Definition:** Combines **WWN zoning** and **Port zoning** in a single SAN.
- **How it works:**
  - Can set some zones based on WWN and others based on port, depending on requirements.
  - Example: Critical servers use WWN zoning for flexibility, and fixed infrastructure uses port zoning.
- **Advantages:**
  - Balances flexibility and security.
  - Useful in complex SANs with varied requirements.
- **Disadvantages:**
  - More complex to manage than using only one zoning type.

Comparison Table

Type	Based On	Cabling Change Impact	Flexibility	Management Complexity
WWN Zoning	Device WWN	No impact	High	Medium
Port Zoning	Switch Port	Needs reconfiguration	Low	Low
Mixed Zoning	WWN + Port	Partial impact	Medium	High